AMR Atlas - Antibiotic Resistance Report

Research and Educational Use Only - Not for Clinical Decision-Making

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Patient Information

Patient ID	P001
Age	34
Sex	Female
Weight (kg)	62.0
Pregnant	No
Age Group	Adult
Comorbidities	Diabetes
Known Allergies	penicillin
Infection Site	Throat

Isolate Information

Species	Streptococcus pneumoniae
Isolation Type	Clinical
Country	USA
Collection Year	2024

Detected Resistance Genes:

gene_mef(a), gene_msr(d), gene_tet(m), gene_pbp2x

Ranked Antibiotic Recommendations

Rank	Antibiotic	Drug Class	Resist. %	Effect. %	Score	Recommendation
1	Clindamycin	Lincosamide	10.3%	89.7%	0.90	Recommended
2	Meropenem	Carbapenem	21.2%	78.8%	0.67	Use with Caution
3	Ceftriaxone	Cephalosporin	21.6%	78.4%	0.67	Use with Caution
4	Erythromycin	Macrolide	46.3%	53.7%	0.54	Use with Caution
5	Tetracycline	Tetracycline	47.1%	52.9%	0.53	Use with Caution
6	Tmp_smx	Sulfonamide	48.8%	51.2%	0.51	Use with Caution
7	Amoxicillin	Beta-lactam (Penicillin)	20.3%	79.7%	0.00	Contraindicated (Allergy)
8	Penicillin	Beta-lactam (Penicillin)	22.1%	77.9%	0.00	Contraindicated (Allergy)

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Methodology Summary

Resistance probabilities were predicted using a soft-voting ensemble of six machine learning classifiers (Logistic Regression, Random Forest, SVM, KNN, XGBoost, MLP) trained on genomic and metadata features from 529 Streptococcus isolates (S. pyogenes, S. pneumoniae, S. agalactiae) sourced from NCBI Pathogen Detection. Each antibiotic-specific model was validated using 5-fold stratified cross-validation, with SMOTE applied only within training folds.

Final ranking scores combine the model's predicted effectiveness (1 - P(resistant)) with patient-specific adjustment factors for known allergies, pregnancy, age group, and infection site, based on clinical literature cited in the accompanying paper.

Disclaimer

This report is generated for research and educational purposes only. It is not a substitute for clinical diagnosis, laboratory antimicrobial susceptibility testing, or professional medical advice. Predictions are based on a research dataset and machine learning models that have not been validated for clinical use. Always consult a qualified healthcare professional and confirmed laboratory AST results before making treatment decisions. No patient data is stored - all inputs are processed in real time and discarded.